Finite length of ordinary regret matching under cyclic and random activation

Abstract

Ordinary regret matching retains signed cumulative regrets, whose negative coordinates can postpone strategy changes for arbitrarily long periods. We prove finite total length of the current strategies and of the positive regret states in every finite exact potential game under two sequential activation rules. The cyclic zero-history result follows from a signed-memory potential budget and a weighted gradient estimate at actual cycle endpoints. For random activation, each player's conditional selection probability may depend on the observed history but must have a uniform positive lower bound. The strategy and positive-state lengths are then almost surely finite, even with arbitrary finite deterministic signed history and without a bound on inter-activation gaps. The new estimate charges a pre-update weighted variance to the potential gain of a hypothetical activation. A fixed analytic lift, conditioning on finitely many possible limiting levels, a posterior probability bound, and a first-exit stopping argument yield summable square-root gains. The limit is a single Nash equilibrium, identified through the original signed recurrence. We also treat boundedly fair deterministic schedules and unrelated initial strategy and regret states after a finite alignment prefix. No assertion is made for every merely fair deterministic schedule, for bounded signed regrets, or for uniform iteration complexity.

Keywords: regret matching; potential games; random activation; finite length; Nash equilibrium; analytic gradient inequality

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1 The recurrence and the convergence statements

Let a finite game have players $i = 1, \dots, n$ and action sets of sizes $m_i \geq 2$. Write

$$X = \prod_{i=1}^{n} \Delta_{m_i}, \qquad \Delta_m = \{x \in \mathbb{R}^m : x \geq 0,\ \langle \mathbf{1}, x \rangle = 1\}, \qquad m_* = \max_i m_i. \tag{1}$$

A player with one action may be removed. For $x \in X$, let $u_i(x_{-i}) \in \mathbb{R}^{m_i}$ be player i's full expected utility vector. Assume that the game has an exact potential, with multilinear extension Φ, so that

$$\Phi(z_i, x_{-i}) - \Phi(x_i, x_{-i}) = \langle z_i - x_i, u_i(x_{-i}) \rangle \qquad (z_i \in \Delta_{m_i}). \tag{2}$$

All payoff and potential entries are finite. Set $\Delta_\Phi = \max_X \Phi - \min_X \Phi$.

Time t counts individual activations. At time t, exactly one player $I_t = i$ uses the current profile, including every earlier activation, and performs

$$g_i^t = u_i(x_{-i}^t) - \langle x_i^t, u_i(x_{-i}^t)\rangle \mathbf{1}, \qquad r_i^{t+1} = r_i^t + g_i^t, \tag{3}$$

$$\theta_i^{t+1} = [r_i^{t+1}]_+, \qquad x_i^{t+1} = \begin{cases} \theta_i^{t+1}/\|\theta_i^{t+1}\|_1, & \theta_i^{t+1} \neq 0, \\ x_i^t, & \theta_i^{t+1} = 0. \end{cases} \tag{4}$$

Every other block remains unchanged, and $\theta_i^t = [r_i^t]_+$ at every state time. The signed state r_i is never truncated or reset. In particular, this is ordinary regret matching (RM), rather than RM+, which replaces the signed state by its positive part after each increment. The retention rule at zero positive state is part of the recurrence.

A deterministic initial pair (x^0, r^0) is *compatible* if

$$[r_i^0]_+ \neq 0 \quad \Longrightarrow \quad x_i^0 = \frac{[r_i^0]_+}{\|[r_i^0]_+\|_1}. \tag{5}$$

There is no restriction on x_i^0 when $[r_i^0]_+ = 0$. Compatibility is preserved by (3)–(4). All norms without subscripts below are Euclidean product norms.

Theorem 1 (Cyclic activation from zero history)**.** *Suppose $r_i^0 = 0$ for every player, $x^0 \in X$ is arbitrary, and players are activated in the fixed cyclic order $1, \dots, n$. Then $\Phi(x^t)$ and each $\|\theta_i^t\|_2$ are nondecreasing. At every activation time,*

$$\sum_{i=1}^{n} \frac{\|\theta_i^t\|_2}{\sqrt{m_i}} \leq \Phi(x^t) - \Phi(x^0) \leq \Delta_\Phi, \tag{6}$$

while

$$\sum_{t\geq 0} \|\theta^{t+1} - \theta^t\|_2^2 \leq (\Delta_\Phi)^2 \sum_{i=1}^{n} m_i. \tag{7}$$

Moreover,

$$\sum_{t\geq 0} \|x^{t+1} - x^t\|_2 < \infty, \qquad \sum_{t\geq 0} \|\theta^{t+1} - \theta^t\|_2 < \infty. \tag{8}$$

The full sequence of current profiles converges to one Nash equilibrium. In particular, all cycle endpoints and intermediate activation profiles have the same limit.

For random activation, work on a probability space with a filtration $(\mathcal{F}_t)_{t\geq 0}$. The state (x^t, r^t) and all hypothetical one-player updates at time t are $\mathcal{F}_t$-measurable; I_t and the next state are $\mathcal{F}_{t+1}$-measurable. Assume that for a deterministic $p_* > 0$,

$$p_i^t := \mathbb{P}(I_t = i \mid \mathcal{F}_t) \geq p_* \quad \text{almost surely for every } i, t. \tag{9}$$

The conditional law may depend on the entire observed history. Neither independence nor an identical distribution is assumed.

Theorem 2 (Random activation with signed history)**.** *For every finite deterministic compatible initial pair, the recurrence under* (9) *satisfies* (8) *almost surely and converges almost surely to a single Nash equilibrium. With*

$$D_t = \Phi(x^{t+1}) - \Phi(x^t), \tag{10}$$

one also has

$$D_t \geq 0 \quad \textit{for every } t, \qquad \sum_{t\geq 0} \sqrt{D_t} < \infty \quad \textit{almost surely.} \tag{11}$$

Potential and positive-state norms are nondecreasing pathwise. The schedule need not have bounded inter-activation gaps.

Theorem 2 includes every initial profile with zero signed history. Section 6 proves the analogous finite-length conclusion for deterministic schedules whose every window of some fixed length contains all players. Section 7 removes initial compatibility for both classes of schedules, without claiming monotonicity before alignment.

The theorems concern current full-information strategies. They do not concern empirical play, simultaneous updates, delayed utilities, or sampled feedback. Almost-sure finite length is not a finite expected-length bound. The full signed state may be unbounded even along a constant equilibrium trajectory.

1.1 The source question and the method

The official ICLR 2026 paper of Anagnostides et al. [1] asks in Section 5 whether ordinary RM converges asymptotically even under alternating updates. Its Algorithm 1 and Section 2.2 give the equivalent output-before-feedback convention. Theorem 1 answers that question for the cyclic zero-history recurrence and strengthens point convergence to finite length. Its qualitative conclusion is consistent with the exponential transient for ordinary RM in [1, Corollary 4.5].

Potential and state-norm accounting for RM+ provides methodological ancestry in [1, 2]. The historical preprint [2] must be distinguished from the ICLR paper: its official arXiv record was administratively withdrawn on 24 September 2026 because authorship and affiliation could not be verified. The retained version is cited solely to acknowledge method provenance, not as an unwithdrawn theorem source. That administrative status is not by itself a mathematical refutation. All recurrence-specific estimates used here are proved below, and no theorem from [2] is invoked.

The signed recurrence introduces a nonnegative charge when a coordinate emerges from negative memory. A post-update variance estimate transports that charge to a common cycle endpoint. The random theorem instead requires a pre-update estimate for every hypothetical activation at the current state. It converts the conditional expected

potential gain into control of the current lifted gradient. Negative initial history also requires restricting the analytic lift to players whose positive state eventually becomes nonzero.

The analytic input is the classical gradient inequality, for example [3, Lemma 2.1]. Stochastic uses of the Kurdyka–Łojasiewicz property and conditioning already have a substantial framework; see Fest, Repetti, and Chouzenoux [4]. We do not claim a new general stochastic gradient inequality. We give the posterior and stopping arguments specifically needed here, rather than importing an unchecked stochastic convergence theorem. Earlier no-regret and fictitious-play comparisons such as [5] concern different state variables; the target here is the actual strategy in (4). An exhaustive novelty claim is not made.

2 A potential budget for arbitrary signed memory

At one compatible activation suppress the player index and write

$$\begin{gathered} r' = r + g, \quad \theta = [r]_+, \quad \theta' = [r']_+, \quad d = \theta' - \theta, \\ R = \|\theta\|_2, \quad R' = \|\theta'\|_2, \quad s = \|\theta\|_1, \quad s' = \|\theta'\|_1, \quad D = \langle x' - x, u \rangle. \end{gathered} \tag{12}$$

Compatibility and centering imply $\langle \theta, g \rangle = 0$, also when $\theta = 0$.

Lemma 1 (The signed-memory charge)**.** *Let*

$$B = \{a : r_a \le 0 < r'_a\}, \qquad Z = \sum_{a \in B} (-r_a)\theta'_a \ge 0. \tag{13}$$

Then

$$\langle \theta, d \rangle \ge 0, \qquad R'^2 - R^2 \ge \|d\|_2^2, \tag{14}$$

$$\langle \theta', g \rangle = \langle \theta', d \rangle + Z. \tag{15}$$

If $s' > 0$, then

$$s'D = \frac{R'^2 - R^2 + \|d\|_2^2}{2} + Z \ge \|d\|_2^2 + Z. \tag{16}$$

If $s' = 0$, then $\theta = \theta' = 0$ and $x' = x$. In all cases,

$$D \ge 0, \qquad R' - R \le \sqrt{m}\, D, \qquad \|d\|_2^2 \le s'D. \tag{17}$$

Proof. On an old positive coordinate, $d_a = \max(0, \theta_a + g_a) - \theta_a \ge g_a$. Multiplying by θ_a and summing proves $\langle \theta, d \rangle \ge 0$, and expansion gives (14). On surviving positive coordinates, $g_a = d_a$. On newly positive coordinates, $d_a = \theta'_a = r_a + g_a$, so $g_a - d_a = -r_a$. Coordinates with $\theta'_a = 0$ do not contribute. This proves (15).

If $s' > 0$, then $s'D = \langle \theta', g \rangle$ because $\langle x, g \rangle = 0$. The polarization identity gives (16). Moreover,

$$s'D \ge \langle \theta', \theta' - \theta \rangle \ge R'(R' - R).$$

Since $s' \le \sqrt{m}R'$, the middle inequality in (17) follows. If $s' = 0$, (14) forces $R = 0$; retention gives $D = 0$. No division by a zero state is used. ∎

By exactness, D is the potential gain of the unilateral update. Therefore every compatible activation sequence, with no scheduling restriction, satisfies

$$\sum_i \frac{\|\theta_i^t\|_2 - \|\theta_i^0\|_2}{\sqrt{m_i}} \le \Phi(x^t) - \Phi(x^0) \le \Delta_\Phi. \tag{18}$$

In particular, $\sum_t D_t \le \Delta_\Phi$ and

$$\|\theta_i^t\|_2 \le R_i^0 + \sqrt{m_i}\Delta_\Phi, \quad \|\theta_i^t\|_1 \le W, \quad W := \max_i\{\sqrt{m_i}R_i^0 + m_i\Delta_\Phi\}, \tag{19}$$

where $R_i^0 = \|\theta_i^0\|_2$. The same W bounds the positive one-norm after every *hypothetical* activation at the current state: append that update to the actual prefix and apply the same total potential budget. This observation will be essential for random activation.

Zero initialization yields (6). Telescoping (14) player by player, and using $R_i^t \le \sqrt{m_i}\Delta_\Phi$, gives (7). For zero history one can take $W = m_*\Delta_\Phi$.

If $\Delta_\Phi = 0$, exactness makes every utility vector action-independent and all advantages zero. A compatible trajectory is constant, and every profile is Nash. We henceforth assume $\Delta_\Phi > 0$, so $W > 0$. Fix $G > 0$ bounding the absolute value of every advantage coordinate over X, and $L < \infty$ such that $\nabla\Phi$ is L-Lipschitz on X.

Lemma 2 (Normalization along an initialized path)**.** *For each compatible activation path there is a finite κ such that*

$$\|x^{t+1} - x^t\|_2 \le \kappa\|\theta^{t+1} - \theta^t\|_2 \quad \textit{for every } t. \tag{20}$$

One may take $\kappa = 2\sqrt{m_*}/\delta$, where δ is the minimum of 1 and the first nonzero positive-state norm of each player that ever has one, including initially positive states.

Proof. Positive-state norms are nondecreasing, so the minimum defining δ involves finitely many strictly positive numbers. If none occurs, put $\delta = 1$. For $s, s' > 0$,

$$x' - x = \frac{d}{s'} + \frac{s - s'}{s'}x, \qquad |s - s'| \le \sqrt{m}\|d\|_2, \qquad s' \ge \delta.$$

The bound follows because $\|x\|_2 \le 1$ and $1 + \sqrt{m} \le 2\sqrt{m}$. At a zero-to-positive transition, $\|d\|_2 \ge \delta$ and the simplex diameter is $\sqrt{2}$. At a zero-to-zero transition both movements vanish. ∎

The constant δ may depend on the entire realized path. No uniform positive lower bound on played probabilities, or on the first positive norm, is assumed.

Lemma 3 (Identification of a convergent fair trajectory)**.** *Suppose a compatible trajectory converges in strategies, each player's positive state is bounded, and every player is activated infinitely often. Its limit is Nash.*

Proof. If action a of player i has limiting advantage $c > 0$, continuity makes its advantage at least $c/2$ at every sufficiently late activation of i. The original signed update then sends $r_i[a]$ to $+\infty$, regardless of its finite earlier negative history. This contradicts boundedness of $[r_i]_+$. All limiting advantages are nonpositive, exactly the Nash condition. ∎

3 Two weighted variance estimates

Write

$$\mathrm{Var}_x(u) = \sum_a x_a\big(u_a - \langle x, u\rangle\big)^2.$$

We need separate estimates at the post-update and pre-update strategy. Their roles in the two scheduling arguments are different.

3.1 The post-update estimate and the zero-history invariant

If a player starts from zero signed history and its positive state is still zero, its strategy has remained x_i^0. Every increment has zero inner product with that fixed strategy. Hence

$$\langle x_i^0, r_i\rangle = 0, \qquad r_i[a] = 0 \quad (a \in \operatorname{supp} x_i^0) \quad \text{while } \theta_i = 0. \tag{21}$$

The second assertion follows because all regret coordinates are then nonpositive. At a zero-to-zero activation, the same support identity before and after the update implies $g_i[a] = 0$ on the retained support.

Lemma 4 (Post-update variance)**.** *At a compatible activation with $s' > 0$,*

$$\mathrm{Var}_{x'}(u) \le (W + G)D. \tag{22}$$

The estimate also holds at zero-to-zero activations under zero signed initialization.

Proof. On the support of θ' outside the newly positive set B in (13), $g_a = d_a$. Thus

$$\sum_{a \notin B} x'_a g_a^2 \le \|d\|_2^2 \le s'D \le WD.$$

On B, $0 < \theta'_a = d_a \le g_a \le G$, and

$$\sum_{a \in B} \theta'_a g_a = \sum_{a \in B} d_a^2 + Z \le s'D.$$

Consequently $\sum_{a\in B} x'_a g_a^2 \le GD$. Subtracting the weighted mean $\langle x', g\rangle = D$ can only reduce the second moment. If $s' = 0$ under zero history, (21) makes the variance zero, as is D. ∎

3.2 The pre-update estimate

Lemma 5 (Pre-update variance)**.** *At any compatible activation with $s > 0$,*

$$\mathrm{Var}_x(u) \le CD, \qquad C = \max\{2W, 8m_*G\}. \tag{23}$$

The constant is independent of the magnitude of the negative signed memory and of δ in Lemma 2.

Proof. Partition the old positive coordinates into survivors P_s and coordinates P_k killed by the update; let N be the newly positive coordinates. Subtracting $\langle \theta, g\rangle = 0$ from $\langle \theta', g\rangle = s'D$ gives the exact identity

$$s'D = \sum_{a\in P_s} g_a^2 + \sum_{a\in P_k} \theta_a(-g_a) + \sum_{a\in N} \theta'_a g_a. \tag{24}$$

Every term is nonnegative. On P_s, $\theta_a \le s \le W$; on P_k, $g_a \le -\theta_a < 0$ and $g_a^2 \le G(-g_a)$. With $H = \max(W, G)$,

$$s\,\mathrm{Var}_x(u) = \sum_{\theta_a>0} \theta_a g_a^2 \le Hs'D. \tag{25}$$

If $s' \le 2s$, this proves (23), since $2H \le C$.

If $s' > 2s$, the state-motion budget and Cauchy–Schwarz give

$$D \ge \frac{\|d\|_2^2}{s'} \ge \frac{(s'-s)^2}{ms'} \ge \frac{s'}{4m}. \tag{26}$$

Let $M = \max_{\theta_a>0}[g_a]_+$. Every old positive coordinate with $g_a > 0$ survives and has $\theta'_a = \theta_a + g_a \ge g_a$, so $M \le s'$. Since $\sum_a x_a g_a = 0$,

$$\mathrm{Var}_x(u) \le G\sum_a x_a|g_a| = 2G\sum_a x_a[g_a]_+ \le 2GM \le 2Gs' \le 8mGD. \tag{27}$$

This proves the second case and the lemma. ∎

Under zero history, (23) also holds for hypothetical updates from a zero positive state. Indeed, (21) holds at the current state, so positive support advantages satisfy $g_a = \theta'_a \le s'$. If $s' > 0$, (16) gives $D \ge \|\theta'\|_2^2/s' \ge s'/m$, and (27) gives $\mathrm{Var}_x(u) \le 2mGD$. If $s' = 0$, support advantages are nonpositive with weighted mean zero, and the variance is zero.

This zero-state extension is false with arbitrary signed history. For

$$r = (-10, -10), \qquad x = (1/2, 1/2), \qquad u = (1, -1), \tag{28}$$

the next positive state is zero and $D = 0$, but $\mathrm{Var}_x(u) = 1$. The random proof will freeze players whose positive state never becomes nonzero, rather than assume the missing support identity.

4 A fixed analytic lift and the cyclic theorem

Consider the compact product of full spheres

$$S = \prod_i \mathbb{S}^{m_i - 1}, \qquad F(y) = \Phi(y_1^{\odot 2}, \dots, y_n^{\odot 2}). \tag{29}$$

The function F is analytic on a domain that does not change with the supports. At $x_i = y_i^{\odot 2}$ its Riemannian gradient is

$$\begin{aligned} \operatorname{grad}_i F(y) &= 2(I - y_i y_i^T)\operatorname{diag}(y_i)\nabla_i \Phi(x) \\ &= 2\operatorname{diag}(y_i)\big(\nabla_i \Phi(x) - \langle x_i, \nabla_i \Phi(x)\rangle \mathbf{1}\big). \end{aligned} \tag{30}$$

The linear operator preceding $\nabla_i \Phi$ in the first line has norm at most 2. Exactness makes $u_i(x_{-i})$ and $\nabla_i \Phi(x)$ differ only by an action-independent scalar. Thus, at $y_i = \sqrt{x_i}$,

$$\|\operatorname{grad}_i F(y)\|_2^2 = 4 \operatorname{Var}_{x_i}(u_i(x_{-i})). \tag{31}$$

We never differentiate the inverse square-root map.

Lemma 6 (Analytic level inequalities)**.** *An analytic function f on a compact product of spheres has finitely many critical values. For each such value λ there exist deterministic $K < \infty$, $\eta > 0$, and $\alpha \in [1/2, 1)$ such that*

$$0 < |f(y) - \lambda| < \eta \quad \Longrightarrow \quad |f(y) - \lambda|^\alpha \le K \|\operatorname{grad} f(y)\|_2. \tag{32}$$

If a sequence approaches its compact cluster set, and that set consists of critical points at one value λ, the same kind of inequality holds along its sufficiently late terms.

Proof. The local analytic gradient inequality in Euclidean coordinates is the only analytic theorem used; see [3, Lemma 2.1]. Near any sphere point, select a coordinate in each block with absolute value at least $1/\sqrt{m_i}$. On a smaller neighborhood its sign is fixed, and solving for it as the signed square root of one minus the other squared coordinates gives an analytic chart with bounded derivative. The chain rule bounds the coordinate gradient by a constant times the Riemannian gradient. The local inequality therefore holds on the manifold. Its exponent can be increased to a number in $[1/2, 1)$ after restricting the value difference below 1.

Around any critical point z, the local inequality forces every critical point in that neighborhood to have value $f(z)$. A finite cover of the compact critical set gives finitely many critical values. Fix λ. Cover the critical fiber at λ by finitely many such neighborhoods, taking the largest exponent and a common constant. Choose a closed value band around λ containing no other critical value. Outside the chosen neighborhoods the part of this band is compact and contains no critical point, so its gradient norm has a positive lower bound, unless that complement is empty. Increasing the constant gives (32) on a smaller

band. The assertion about a sequence also follows by a finite cover of its cluster set and compactness. ∎

For a cyclic trajectory write $z^k = x^{kn}$, and let $z^{k,i}$ be the profile after player i in cycle k, so $z^{k,0} = z^k$ and $z^{k,n} = z^{k+1}$. Set

$$D_i^k = \Phi(z^{k,i}) - \Phi(z^{k,i-1}), \quad \gamma_k = \sum_i D_i^k, \quad d_i^k = \theta_i^{(k+1)n} - \theta_i^{kn}. \tag{33}$$

Each block moves once. Lemmas 1 and 2 yield

$$\sum_i \|d_i^k\|_2^2 \le W\gamma_k, \qquad \|z^{k+1} - z^{k,i-1}\|_2^2 \le \kappa^2 W\gamma_k. \tag{34}$$

For the second bound, only blocks $j \ge i$ differ, so the squared product norm is the sum of their squared actual movements.

Lemma 7 (The actual cycle endpoint)**.** *Under zero history, with $y^k = \sqrt{z^k}$,*

$$\|\mathrm{grad}\, F(y^{k+1})\|_2^2 \le C_0^2\gamma_k, \qquad C_0^2 = 8(W + G + nL^2\kappa^2 W). \tag{35}$$

Proof. At the cycle endpoint, block i is its post-activation strategy. Apply Lemma 4 to the utility vector used at that activation, or equivalently to $\nabla_i\Phi(z^{k,i-1})$. Its lifted gradient has norm at most $2\sqrt{(W+G)D_i^k}$. Replacing that vector by $\nabla_i\Phi(z^{k+1})$ changes the lifted block gradient by at most

$$2L\|z^{k+1} - z^{k,i-1}\|_2 \le 2L\kappa\sqrt{W\gamma_k}.$$

Square using $(a+b)^2 \le 2a^2 + 2b^2$ and sum over i. The zero-state case is covered by the zero-history part of Lemma 4. ∎

Lemma 8 (A shifted finite-length estimate)**.** *Suppose $\Phi(z^k)$ is nondecreasing and bounded, with gains γ_k, and an analytic lift satisfies* (35) *for a fixed finite constant. Then*

$$\sum_{k\ge 0} \sqrt{\gamma_k} < \infty. \tag{36}$$

Proof. Write $\Phi(z^k) \uparrow \ell$. Since $\gamma_k \to 0$, every cluster point of y^k is critical and has value ℓ. Lemma 6 provides a fixed $K < \infty$ and $\alpha \in [1/2, 1)$ such that, for all sufficiently large k,

$$e_k^\alpha \le K\|\mathrm{grad}\, F(y^k)\|_2 \le KC_0 a_{k-1}, \qquad e_k = \ell - \Phi(z^k), \quad a_k = \sqrt{\gamma_k}.$$

If some $e_k = 0$, all subsequent gains are zero. Otherwise $e_k > 0$, and in the displayed tail also $a_{k-1} > 0$. Put $\psi(e) = Ke^{1-\alpha}/(1-\alpha)$. Concavity gives

$$\psi(e_k) - \psi(e_{k+1}) \ge Ke_k^{-\alpha}\gamma_k \ge \frac{a_k^2}{C_0 a_{k-1}}.$$

The inequality $2a_k \le a_{k-1} + a_k^2/a_{k-1}$ implies

$$2a_k \le a_{k-1} + C_0\big(\psi(e_k) - \psi(e_{k+1})\big).$$

Summing from a sufficiently large deterministic index K_0 through N yields

$$\sum_{k=K_0}^{N} a_k \le a_{K_0-1} - a_N + C_0\big(\psi(e_{K_0}) - \psi(e_{N+1})\big) \le a_{K_0-1} + C_0\psi(e_{K_0}).$$

This bounds the tail and proves the lemma. ∎

Completion of Theorem 1. The budget and square-sum assertions follow from Section 2. By (34) and Lemma 8,

$$\sum_k \sum_i \|d_i^k\|_2 \le \sqrt{nW} \sum_k \sqrt{\gamma_k} < \infty.$$

This is the positive-state length at the individual-activation scale. Lemma 2 gives the strategy length. The full profile sequence is therefore Cauchy, and Lemma 3 identifies its limit as Nash. The proof controls all intermediate profiles, not only the cycle endpoints. ∎

5 Random activation and conditioning on limiting levels

We now prove Theorem 2. Write D_i^t for the potential gain if player i alone were activated at the state (x^t, r^t). These quantities are $\mathcal{F}_t$-measurable and nonnegative, and $D_t = D_{I_t}^t$. With

$$\mu_t = \mathbb{E}[D_t \mid \mathcal{F}_t] = \sum_i p_i^t D_i^t, \tag{37}$$

Tonelli's theorem and the pathwise budget show

$$\mathbb{E}\sum_{t\ge 0} \mu_t = \mathbb{E}\sum_{t\ge 0} D_t \le \Delta_\Phi, \qquad \sum_{t\ge 0} \mu_t < \infty \quad \text{almost surely.} \tag{38}$$

In particular, $\mu_t \to 0$ almost surely.

5.1 Players that ever have a positive state

Define the terminal set

$$J(\omega) = \{i : \theta_i^t(\omega) \ne 0 \text{ for some finite } t\}. \tag{39}$$

A player in J stays positive after its first nonzero positive state. Every player outside J retains the deterministic strategy x_i^0 forever. On each path all the players in J are positive after a finite time, since there are only finitely many players.

For a deterministic nonempty subset J, define the fixed analytic function

$$F_J(y_J) = \Phi\big((y_i^{\odot 2})_{i\in J}, (x_i^0)_{i\notin J}\big), \qquad S_J = \prod_{i\in J} \mathbb{S}^{m_i-1}. \tag{40}$$

The complement is frozen at deterministic values. On a realized path with terminal set J, after all its positive states have appeared, Lemma 5 and (31) give

$$\|\operatorname{grad} F_J(\sqrt{x_J^t})\|_2^2 \le 4C \sum_{i\in J} D_i^t \le \frac{4C}{p_*}\mu_t. \tag{41}$$

No support stabilization within those players is needed. This is a pathwise comparison with the original adapted μ_t; it does not condition on J at a random entrance time.

Potential monotonicity gives a random limit ℓ. By (38), (41), and compactness, ℓ is a critical value of F_J. Lemma 6 gives finitely many such values. Thus the events

$$A_{J,\lambda} = \{J(\omega) = J,\ \lim_t \Phi(x^t) = \lambda\} \tag{42}$$

form a finite partition of the event $J \neq \varnothing$, up to a null set, when λ ranges over the critical values of F_J. If $J = \varnothing$, the strategies and positive states are constant and every $D_t = 0$. The Nash conclusion for that event will be addressed separately below.

5.2 A posterior bound under the conditioned law

One cannot directly pull $(\ell - \Phi(x^t))^{-\alpha}$ out of a conditional expectation given $\mathcal{F}_t$: the random limit ℓ depends on the future. Fix instead one event $A = A_{J,\lambda}$ with positive probability, and put

$$\mathbb{Q} = \mathbb{P}(\,\cdot \mid A), \qquad h_t = \mathbb{P}(A \mid \mathcal{F}_t).$$

The event A belongs to $\sigma(\bigcup_t \mathcal{F}_t)$, since it is specified by the whole state trajectory. The bounded martingale h_t converges to $\mathbf{1}_A$ under $\mathbb{P}$, and hence to 1 under $\mathbb{Q}$. Also $h_t > 0$ for every t, outside a $\mathbb{Q}$-null set. Conditional Bayes' formula yields

$$\begin{aligned} q_i^t := \mathbb{Q}(I_t = i \mid \mathcal{F}_t) &= \frac{\mathbb{P}(A \cap \{I_t = i\} \mid \mathcal{F}_t)}{h_t} \\ &\ge \frac{p_i^t - (1 - h_t)}{h_t}. \end{aligned} \tag{43}$$

Consequently,

$$h_t > 1 - p_*/2 \quad \Longrightarrow \quad q_i^t \ge p_*/2 \quad \text{for every } i. \tag{44}$$

Conditioning on a terminal event does not leave the original selection law unchanged. The posterior estimate supplies the weaker uniform floor on the eventual tail.

Under $\mathbb{Q}$, the level λ is deterministic, and

$$e_t = \lambda - \Phi(x^t) \ge 0 \tag{45}$$

is adapted, nonincreasing, and tends to zero. Set $m_t = \mathbb{E}_{\mathbb{Q}}[D_t \mid \mathcal{F}_t]$. On the event in (44), once every player in J has a positive state, the first bound in (41) gives

$$m_t = \sum_i q_i^t D_i^t \geq a\|\operatorname{grad} F_J(\sqrt{x_J^t})\|_2^2, \qquad a = \frac{p_*}{8C} > 0. \tag{46}$$

All statements in this paragraph are under the conditioned measure and its unchanged filtration.

5.3 A first-exit argument for the square-root gains

Apply Lemma 6 to the deterministic function F_J and value λ. Choose its deterministic constants K, η, α in (32), and put

$$b = K/\sqrt{a}, \qquad \psi(e) = \frac{e^{1-\alpha}}{1-\alpha}. \tag{47}$$

On the $\mathcal{F}_t$-measurable event

$$G_t = \{h_t > 1 - p_*/2,\ 0 < e_t < \eta,\ \theta_i^t \neq 0 \text{ for all } i \in J\}, \tag{48}$$

equations (32) and (46) imply

$$e_t^\alpha \leq b\sqrt{m_t}. \tag{49}$$

Concavity of ψ, the identity $e_{t+1} = e_t - D_t \geq 0$ under $\mathbb{Q}$, and conditional Jensen give

$$\begin{aligned} \mathbb{E}_{\mathbb{Q}}[\psi(e_t) - \psi(e_{t+1}) \mid \mathcal{F}_t] &\geq e_t^{-\alpha} m_t \\ &\geq b^{-1}\sqrt{m_t} \\ &\geq b^{-1}\mathbb{E}_{\mathbb{Q}}[\sqrt{D_t} \mid \mathcal{F}_t] \qquad \text{on } G_t. \end{aligned} \tag{50}$$

Every division occurs only on $e_t > 0$. Notice that this argument requires a lower bound on the expected gain by the squared gradient, not an upper bound on the realized step by the current gradient.

For each deterministic integer $K_0 \geq 0$, define

$$\tau_{K_0} = \inf\{t \geq K_0 : G_t \text{ fails}\}, \qquad \inf \varnothing = \infty. \tag{51}$$

This is a stopping time. For $t \geq K_0$, the indicator of $t < \tau_{K_0}$ is $\mathcal{F}_t$-measurable. Multiply (50) by that indicator, take expectations, and sum through a finite $N \geq K_0$. The actual stopped differences telescope pathwise, so

$$\begin{aligned} \mathbb{E}_{\mathbb{Q}} \sum_{t=K_0}^{N} \mathbf{1}_{\{t<\tau_{K_0}\}} \sqrt{D_t} &\leq b\, \mathbb{E}_{\mathbb{Q}}\Big[\psi(e_{K_0}) - \psi\big(e_{\min(\tau_{K_0}, N+1)}\big)\Big] \\ &\leq b\, \psi(\Delta_\Phi). \end{aligned} \tag{52}$$

Here $0 \le e_{K_0} \le \Delta_\Phi$ because both the current potential and its limit lie in its global range. Monotone convergence shows that, for each deterministic K_0,

$$\sum_{K_0 \le t < \tau_{K_0}} \sqrt{D_t} < \infty \quad \mathbb{Q}\text{-almost surely.} \tag{53}$$

Intersect these probability-one assertions over the countably many integers K_0.

If $e_t = 0$ at some time, monotonicity makes every later gain zero. Otherwise $e_t > 0$ for all t; since $h_t \to 1$, $e_t \to 0$, and all players in J eventually have positive state, G_t holds forever after some finite time. Such a path belongs to $\{\tau_{K_0} = \infty\}$ for some deterministic integer K_0. On the countable intersection above its entire tail is one of the finite sums in (53). Adding the finite prefix proves $\sum_t \sqrt{D_t} < \infty$ under $\mathbb{Q}$. Taking the finite union over the positive-probability events (42), and adding $J = \varnothing$, proves (11) under $\mathbb{P}$.

This argument uses first exits from measurable good events and countably many deterministic starting times. It does not use a last entrance time as a stopping time, and it makes no uniform integrability claim about such an entrance time.

5.4 Physical finite length and equilibrium identification

The state-motion budget and (19) now give

$$\sum_t \|\theta^{t+1} - \theta^t\|_2 \le \sqrt{W} \sum_t \sqrt{D_t} < \infty \quad \text{almost surely.} \tag{54}$$

Lemma 2 then gives finite strategy length, with a path-dependent finite κ. It does not imply integrability of that length.

Assumption (9) implies that every player is activated infinitely often almost surely. Iterated conditioning bounds the probability of avoiding a fixed player over the next k activations, conditional on the present history, by $(1 - p_*)^k$. Letting $k \to \infty$ and taking a countable union over possible last activation times proves the assertion. The full strategy sequence converges by finite length, so Lemma 3 identifies its limit. This also handles never-positive players and the constant trajectories with $J = \varnothing$. Restricted stationarity alone would not enforce their best-response conditions. Theorem 2 is proved.

6 Boundedly fair deterministic activation

Call a deterministic schedule *boundedly fair* if some integer $B \ge 1$ has the property that every consecutive block of B activations contains every player. Fixed cyclic order is one example, but neither a fixed order nor one update per player per window is required.

Theorem 3 (Boundedly fair activation)**.** *For a finite deterministic compatible initial pair and a boundedly fair schedule, ordinary RM satisfies* (8) *and converges to a single Nash equilibrium. Potential and positive-state norms are nondecreasing. In fact, the square roots of the total potential gains in consecutive windows of length B are summable.*

Proof. The budget and normalization lemmas apply to the entire trajectory. Define the set J of players whose positive state ever becomes nonzero. If J is empty, the trajectory is constant; fairness and Lemma 3 identify it as Nash. Otherwise freeze the complement and use the analytic function F_J from (40). Choose an integer t_0 after all players in J have positive state, and partition the tail into windows $[t_0 + kB, t_0 + (k+1)B)$.

Write γ_k for the total potential gain in window k, and $d_t = \theta^{t+1} - \theta^t$. Then

$$\sum_{t \text{ in window } k} \|d_t\|_2^2 \le W\gamma_k, \qquad \sum_{t \text{ in window } k} \|x^{t+1} - x^t\|_2 \le \kappa\sqrt{BW\gamma_k}. \tag{55}$$

For each $i \in J$, choose its last activation s_i in that window. Its post-update strategy is precisely block i at the endpoint $z^{k+1} = x^{t_0+(k+1)B}$. Because its state is positive, Lemma 4 applies without the zero-history invariant. The lifted block gradient using $\nabla_i\Phi(x^{s_i})$ has norm at most $2\sqrt{(W+G)D_{s_i}}$. Transporting this utility vector to the window endpoint changes it by at most $2L\kappa\sqrt{BW\gamma_k}$.

The last-activation gains D_{s_i} form a subset of the nonnegative gains in the window. Squaring, summing over $i \in J$, and writing $y_J^{k+1} = \sqrt{z_J^{k+1}}$ therefore gives

$$\|\operatorname{grad} F_J(y_J^{k+1})\|_2^2 \le 8\big(W + G + |J|L^2\kappa^2 BW\big)\gamma_k. \tag{56}$$

The proof of Lemma 8, now for F_J and these endpoints, gives $\sum_k \sqrt{\gamma_k} < \infty$. Equation (55) gives finite strategy length. Similarly,

$$\sum_{t \text{ in window } k} \|d_t\|_2 \le \sqrt{BW\gamma_k},$$

so the positive-state length is finite. A finite prefix has no effect. Bounded fairness supplies infinitely many activations of every player, and Lemma 3 finishes the proof. ∎

Theorem 2 is not a bounded-window corollary. For independent sampling with replacement and at least two players, every player's inter-activation gaps are almost surely unbounded. For any fixed gap length, disjoint blocks that omit that player are independent events of the same positive probability, hence occur infinitely often; intersect over the positive integers. The conditional-gain proof applies without choosing any finite random window bound.

7 Independently specified initial states

Initial compatibility is needed for the one-step identity $\langle \theta, g \rangle = 0$, not for eventual convergence itself.

Corollary 7.1 (Finite alignment prefix)**.** *Let (x^0, r^0) be any finite deterministic pair, not necessarily compatible. Under either* (9) *or a boundedly fair deterministic schedule, the strategy and positive-state lengths in* (8) *are finite, almost surely in the random case, and*

the current profile converges to a single Nash equilibrium. The monotone-potential and positive-state budget conclusions apply after every player has been activated at least once, and need not hold before then.

Proof. Immediately after a player's first activation, the output rule makes its state compatible: it either normalizes a nonzero positive state or has zero positive state, when compatibility is vacuous. Compatibility is then preserved. For boundedly fair schedules, every player has been activated by time B, so Theorem 3 applies to the deterministic compatible state at that time.

For random activation, the first cover time

$$T = \inf\{k \geq 1 : \{I_0, \dots, I_{k-1}\} = \{1, \dots, n\}\}$$

is finite almost surely by (9). Let $\mathcal{W}$ be the countable set of finite player words that first cover all players at their last letter. For $w = (w_0, \dots, w_{k-1}) \in \mathcal{W}$, put

$$B_w = \{I_0 = w_0, \dots, I_{k-1} = w_{k-1}\} \in \mathcal{F}_k.$$

These disjoint events cover $\{T < \infty\}$. The state at time k is a fixed deterministic compatible pair on B_w: the initial pair is deterministic and the state transitions depend only on the selected players. Each finite signed state is finite because all increments are bounded.

If $\mathbb{P}(B_w) > 0$, condition on B_w. For $t \geq k$, this event is already in $\mathcal{F}_t$, so under $\mathbb{P}_w = \mathbb{P}(\,\cdot \mid B_w)$,

$$\mathbb{P}_w(I_t = i \mid \mathcal{F}_t) = \mathbb{P}(I_t = i \mid \mathcal{F}_t) \geq p_* \quad \mathbb{P}_w\text{-almost surely.}$$

Theorem 2 applies to the shifted filtration and the deterministic state on B_w. Any other information in $\mathcal{F}_k$ is harmless; the theorem does not require a trivial initial sigma-field. The tail has finite length and a Nash limit. Taking the countable union over words and adding each finite prefix proves the assertion. Unlike terminal-event conditioning in Section 5, conditioning on a completed prefix genuinely preserves the later conditional selection probabilities. ∎

Pre-alignment monotonicity can fail. In the one-player game $u = (1, 0)$, take $x^0 = (1, 0)$ and $r^0 = (0, 2)$. The first advantage is $(0, -1)$, hence $r^1 = (0, 1)$ and $x^1 = (0, 1)$. The potential decreases by 1. Thus the expression $\sqrt{D_t}$ is not asserted from time zero in Corollary 7.1, although the two physical finite-length conclusions are.

8 Scope and obstructions

The distinction between signed regrets and their positive parts cannot be removed. In a one-player two-action game with $u = (1, 0)$, $x^0 = (1, 0)$, and $r^0 = 0$, every profile equals

x^0, whereas $r^t = (0, -t)$. Positive states and strategies have zero length, but the signed state is unbounded.

Arbitrary negative history also creates arbitrarily long initial delays even in a fixed one-player game. With $u = (1, -1)$, $x^0 = (1/2, 1/2)$, and $r^0 = (-M, -M)$ for an integer $M \geq 1$, the first M activations leave the strategy unchanged. The $(M + 1)$st activation creates a positive first coordinate and moves to the optimal pure strategy. This illustrates both the zero-state obstruction in (28) and the absence of a convergence-time bound independent of the initial history.

The random conclusion excludes exceptional schedules and does not prove convergence for every deterministic schedule in which each player appears infinitely often. Under that weaker fairness condition there is no bounded transport window, nor a conditional lower bound that captures a fixed fraction of the current hypothetical gains. Whether ordinary RM has finite length or even converges for every such schedule is not settled here. Nor is the theorem extended to conditional activation probabilities that tend to zero merely with a divergent cumulative sum.

All proofs require sequential updates using exact current utility vectors. Simultaneous moves, stale or delayed information, and noisy or bandit feedback alter the potential identity and are outside the conclusions. The present finite critical-value argument uses deterministic initial profiles; no general random-initialization theorem is asserted.

The constants arising from normalization, analytic exponents, and entry into a suitable level band need not be effective uniformly over games and initialized paths. Neither a uniform iteration complexity nor a finite expected strategy length follows. The auxiliary sequence $\sqrt{x^t}$ lives in a useful compact analytic domain, but its finite length is neither needed nor claimed.

Revision note

This revised research manuscript retains the cyclic zero-history theorem and its signed-memory proof. Its additional results are the pre-update variance estimate, almost-sure finite length under history-dependent uniformly exploratory random activation, arbitrary compatible deterministic signed history, the boundedly fair deterministic extension, and the finite-alignment corollary. The conditioning and stopping proof is supplied in full. The historical source status of [2] has been corrected, while its methodological provenance is retained. Novelty beyond the specifically compared sources has not been fully established; this is a research revision rather than a submission-ready final version.

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